ALUMINUM / PLASTIC								R (Rough) F (Finish)	R (Rough) F (Finish)	Adaptive (1/2 of tool diameter) Slotting (1/4 of tool diameter) Contour Finishing (1 1/2 of tool diameter)		
Material	**Operation**	**Tool**	**SFM**	**Speed (rpm)**	**Feed (ipm)**	**Flute**	**FPT**	**Stepover**	**Helix/Ramp**	**Stepdown**	**Confidence/Notes**	**Notes**
Aluminum	Face Finish	1/4" endmill	523	8000	30	3	0.0013	0.1		0.01		Face Finish (tool diameter x 40%)
Aluminum	Adaptive	1/4" endmill	523	8000	30	3	0.0013	R - 0.02 F - 0.01	R - 2 deg F - 3 deg	0.125		
Aluminum	Slotting (Contour - Full Width)	1/4" endmill	392	6000	7	3	0.0004		1 deg	0.0625		
Aluminum	Contour Finishing	1/4" endmill	392	6000	10	3	0.0006	0.01		0.375		
Aluminum	Bore (Helix)	1/4" endmill	523	8000	30	3	0.0013	R - 0.02 F - 0.01	R - 2 deg F - 3 deg			
Aluminum	Face Finish	3/16" endmill	392	8000	25	3	0.0010	0.075		0.01		
Aluminum	Adaptive Roughing	3/16" endmill	392	8000	25	3	0.0010	R - 0.018 F - 0.01	R - 0.018 F - 0.01	0.093		
Aluminum	Slotting (Contour - Full Width)	3/16" endmill	294	6000	7	3	0.0004		0.9 deg	0.0468		
Aluminum	Contour Finishing	3/16" endmill	294	6000	10	3	0.0006	0.008		0.2812		
Aluminum	Bore (Helix)	3/16" endmill	327	8000	25	3	0.0010	R - 0.018 F - 0.01	R - 0.9 deg F - 1.8 deg			
Aluminum	Face Finish	5/32" endmill	327	8000	25	3	0.0010	0.0625		0.01		
Aluminum	Adaptive Roughing	5/32" endmill	327	8000	25	3	0.0010	R - 0.016 F - 0.01	R - 0.8 deg F - 1.6 deg	0.0785		
Aluminum	Slotting (Contour - Full Width)	5/32" endmill	255	6250	7	3	0.0004		0.8 deg	0.039		
Aluminum	Contour Finishing	5/32" endmill	255	6250	10	3	0.0005	0.007		0.2343		
Aluminum	Bore (Helix)	5/32" endmill	327	8000	25	3	0.0010	R - 0.016 F - 0.01	R - 0.8 deg F - 1.6 deg			
Aluminum	Face Finish	1/8" endmill	262	8000	20	3	0.0008	0.05		0.007		
Aluminum	Adaptive Roughing	1/8" endmill	262	8000	20	3	0.0008	R - 0.014 F - 0.008	R - 0.7 deg F - 1.4 deg	0.0625		
Aluminum	Slotting (Contour - Full Width)	1/8" endmill	212	6500	5	3	0.0003		0.7 deg	0.032		
Aluminum	Contour Finishing	1/8" endmill	212	6500	7	3	0.0004	0.006		0.1875		
Aluminum	Bore (Helix)	1/8" endmill	262	8000	20	3	0.0008	R - 0.014 F - 0.008	R - 0.7 deg F - 1.4 deg			
Aluminum	Face Finish	1/16" endmill	131	8000	15	3	0.0006	0.025		0.006		
Aluminum	Adaptive Roughing	1/16" endmill	131	8000	15	3	0.0006	R - 0.012 F - 0.006	R - 0.6 deg F - 1.2 deg	0.0312		
Aluminum	Slotting (Contour - Full Width)	1/16" endmill	110	6750	5	3	0.0002		0.6 deg	0.0156		
Aluminum	Contour Finishing	1/16" endmill	110	6750	7	3	0.0003	0.005		0.0937		
Aluminum	Bore (Helix)	1/16" endmill	131	8000	10	3	0.0004	R - 0.012 F - 0.006	R - 0.6 deg F - 1.2 deg			
Aluminum	Drill (pecking)	.125 Drill	196	6000	7	2	0.0006			.03 pecks		
Aluminum	Surfacing	1/8" ballmill		8000	10	3	0.0004	0.005		0.005		
Aluminum	Chamfering	1/8" chamfer		8000	20	2	0.0013	0.03 (tip offset)		0.03		

Aluminum	Threadmilling	Threadmills - misc. cutter diameters		8000	10	4	0.0005	0.005				